# MICAH OEVERMANN, PHD


Texas A&M University ◇ College Station, Texas ◇ U.S. Citizen
mjooevermann@gmail.com ◇ (281) 925-9569
micah-oevermann.com ◇ Google Scholar Link


## EDUCATION

**PhD in Mechanical Engineering** *December 2025*
Texas A&M University, College Station, USA
Dissertation Title: Design, Modeling, and Nutational Instabilities of Soft Pendulum Driven Spherical Robots

**B.S. in Mechanical Engineering** *December 2021*
Texas A&M University, College Station, USA

**Study Abroad - Engineering Mechanics** *Summer 2019*
Arts et Métiers ParisTech, Aix-en-Provence, France

## EXPERIENCE

**Robotics and Automation Design Lab** January 2022 - Present
*Graduate Research Assistant* *College Station, TX*

- Led a small team of grad students and engineers on the full-stack development of the RoboBall II prototypes from CAD to concrete
- Pioneered a novel outer shell manufacturing design, control, and pydrake simulation environments for RoboBall
- Led teams of 5, 3, and 2 undergraduates in the three summer programs, all resulting in 3 conference and a journal publications
- Collaborated with the Robotic Space Simulator team on the motion control two 7-dof Stewart platforms with UR-20 robotic arms
- Collaborated with external partners at Southwest Research Institute for the development of motion planning libraries

**BakerRisk Engineering Consultants** August 2020 - December 2020
*Student Co-op, Blast Testing Group* *San Antonio, TX*

- Destructive full-scale structural testing with Deflagration, Vapor Cloud, and Shock Tube methods
- Manufactured mounts for window specimens and piezoelectric pressure or force instrumentation
- Prioritized safety with no major injuries while working around debris fields with broken glass

**Biomechanical Environments Laboratory** January 2019 - May 2019
*Undergrad Research Assistant* *College Station, TX*

- Applied concepts of linear elastic theory in the development of a biaxial tissue testing platform
- Implemented the use of a novel fish hook – line technique on organic samples to reduce clamp stresses
- Presented final poster in a public research symposium

## TEACHING AND COUNSELING EXPERIENCE

**Conference Paper Reviewer** March 2025-Present

- UR 2025 and ICRA 2026 Conferences

**Guest Speaker - MEEN 612** December 2024
*Introduction to Robotic Manipulators* *College Station, TX*

- Lecture Title: *Introduction to Simulating Robot Arms in pyDrake*

**Guest Speaker - MEEN 689** December 2024
*Intuitive and Counterintuitive Mechanisms* *College Station, TX*

- Lecture Title: *Designing for Assembly: Lessons from RoboBall II Pendulum*

**Grad Camp Counselor** Fall 2022
*Graduate Orientation Camp* *College Station, TX*
- Assist incoming graduate students at introductory sessions to the university

**Numerical Methods Helpdesk** August 2019 - December 2019
*Student Worker for MEEN 357* *College Station, TX*
- Assist students with Python code portions of projects and homework

**Volunteer Teaching Assistant** August 2018 - December 2018
*Kinesiology 199 - Whitewater Kayaking* *College Station, TX*
- One-on-one instruction and feedback on fundamental paddling techniques

**Python Teaching Assistant** August 2018 - December 2018
*Student Assistant for Intro Engineering Course ENGR 102* *College Station, TX*
- Assisted in the instruction of freshman engineering students with the basics of coding in Python

## PATENT APPLICATIONS

Interior Molding Method for Construction of configurable Bead Locked Spherical Tires
U.S. Provisional Application No.: 63/940,051 *Filing Date: December 13, 2025*

## HONORS AND AWARDS

**Best Presentation Award** *September 2023*
OSU International Mechatronics Conference and Exposition, 2023

**Senior Capstone: Best in MEEN** *December 2021*
Texas A&M University, College Station, USA

## TECHNICAL SKILL BUZZWORDS

**Mechanical:** SolidWorks ◇ Rapid Prototyping ◇ Machining ◇ Mechanism Design ◇ GD&T ◇ Dynamic Analysis ◇ Molding ◇ Hermetic Seals
**Software:** C++ ◇ Python ◇ MATLAB ◇ ROS2 ◇ Drake ◇ Linux ◇ Git/Github ◇ Docker ◇ LLM Support ◇ Energy Methods ◇ LaTeX
**Electrical:** Soldering ◇ Oscilloscopes ◇ Sensor Integration ◇ Cable Harness ◇ CAN/Ethernet Bus ◇ Battery Management Systems
**Controls:** PID ◇ Disturbance Rejection ◇ LQR ◇ Full-State Feedback Motion Planning ◇ State-Space Control ◇ Underactuated Systems